ProprioVLA: Self-Supervised Proprioceptive Grounding for Vision-Language-Action Policies

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Abstract

Vision-language-action (VLA) policies have shown strong promise as generalist controllers for robot manipulation, yet their action predictions can remain fragile under changes in visual perspective. We identify a systematic failure pattern that lateral camera shifts induce co-directional shifts in predicted robot trajectories, suggesting that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. To bridge this gap, we introduce PROPRIOVLA, a self-supervised alignment approach to incentivize self-grounding capability in VLA policies. PROPRIOVLA guides the policy to localize its own embodiment by aligning learned grounding queries with state-aligned residual flow targets mined from robot interaction. Specifically, PROPRIOVLA turns proprioceptively aligned visual change, captured by optical flow and image difference, into token-level self-grounding targets. It then trains a lightweight learned-query head over image-token embeddings extracted from the VLM's final hidden states to recover these targets with normalized KL supervision, regularizing the visual representation to encode the visible embodiment without manual masks, external grounding datasets, or stronger foundation-model supervision. Extensive experiments on both real-robot and simulation environments show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, suggesting that proprioceptive grounding is a practical route toward spatially reliable VLA control.

1 Introduction

Vision-language-action (VLA) policies have emerged as a promising paradigm for generalist robot manipulation, scaling vision-language pretraining with robot demonstrations to produce language-conditioned controllers across tasks and embodiments [6, 4, 20, 10, 2, 7]. Yet, the spatial representations that support reliable closed-loop control remain insufficiently understood. Recent work has begun to address this issue by injecting 3D structure, spatial action representations, point-cloud or depth cues, or external spatial supervision into VLAs [25, 18, 12, 14, 19, 13]. More broadly, representation supervision has shown promise for improving visual and spatial representations in multimodal and generative models [21, 22, 9]. However, these approaches primarily supervise scene- or object-level spatial understanding, often through external teachers or explicit 3D signals. In contrast, a more basic requirement for robot control remains underexplored: whether a VLA visually grounds the robot body that executes the action.

In this work, we identify a systematic failure pattern suggesting that current VLAs can rely on image-space shortcuts rather than stable robot-scene geometry. We laterally shift the wrist-camera observation while keeping the physical robot-scene state, language instruction, proprioceptive state, and ground-truth action fixed. A spatially grounded policy should preserve the same robot-centric action under this intervention. Instead, we observe that predicted trajectories can drift in the same

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direction as the image shift, indicating sensitivity to pixel-coordinate correlations or camera-specific object-location priors. We further probe the visual representation with robot-centric queries and find that the model often fails to localize the visible gripper or end-effector, even when it recognizes the task-relevant object. This suggests a missing capability we call *visual self-grounding*: binding proprioceptive state to visual evidence of the robot's own embodiment.

To address this gap, we introduce PROPRIOVLA, a self-supervised alignment approach to incentivize self-grounding capability in VLA policies. PROPRIOVLA guides the policy to localize its own embodiment by aligning learned grounding queries with state-aligned residual flow targets mined from robot interaction. The core insight is simple: when only the robot is actuated, visual change aligned with proprioceptive change reveals the robot itself. This makes it possible to derive self-grounding supervision directly from robot demonstrations, without manual masks, external grounding datasets, or stronger foundation-model supervision.

To operationalize this insight, PROPRIOVLA first constructs State-Aligned Residual Flow (SARF) targets from demonstration trajectories. SARF combines optical flow and image differences with proprioceptive state changes to identify residual visual motion likely caused by the robot body, and projects this signal onto the VLA visual-token grid as confidence-weighted self-grounding targets.

Given these targets, PROPRIOVLA trains a lightweight Proprioceptive Alignment Module (PAM) to make the VLA visual tokens recover where the acting body appears. PAM applies learned grounding queries to image-token embeddings from the VLM's final hidden states and predicts a self-grounding distribution, which is matched to the SARF target with a normalized KL objective. Because this branch is auxiliary, SARF flow targets and pseudo-labels are needed only during training; at inference, the policy uses the original VLA action interface unchanged.

We evaluate PROPRIOVLA in both simulation and real-robot manipulation. Across settings, it improves task success and robustness over action-supervised VLA adaptation, reduces view-induced trajectory drift under controlled camera perturbations, and strengthens localization of the visible gripper or end-effector. These results suggest that visual self-grounding is a critical representational capability for spatially reliable VLA control.

Our contributions are summarized as follows:

- We identify a missing-self failure mode in VLA policies: under camera-only lateral perturbations, predicted robot trajectories can drift with the image shift, revealing image-space shortcuts rather than stable robot-centric grounding. We further connect this failure to weak visual self-grounding through interpretable representation probes.
- We introduce PROPRIOVLA, a self-supervised proprioceptive alignment approach that induces visual self-grounding by aligning learned grounding queries with state-aligned residual flow targets mined from robot interaction, without requiring manual masks, external grounding datasets, or stronger foundation-model supervision.
- Experiments in simulation and real-robot manipulation show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, while reducing view-induced trajectory drift and improving localization of the visible gripper or end-effector.

2 Related Work

Vision-language-action models. Vision-language-action (VLA) policies scale language-conditioned robot control by coupling vision-language backbones with robot action prediction. RT-1 and RT-2 established the value of large robot datasets and web-scale semantic priors [3, 5]; recent open and generalist models further improve transfer across tasks, embodiments, and action parameterizations [16, 11, 1]. However, this line of work mainly optimizes task success under the training observation distribution. It does not directly test whether the action-relevant visual evidence is anchored to camera-invariant robot-object geometry. PROPRIOVLA studies this missing failure mode: under camera perturbations, a fine-tuned VLA can follow image-coordinate shortcuts instead of grounding actions in the robot's own state.

Visual grounding in VLA policies. Recent work improves VLA perception by making space, viewpoint, or proprioception more explicit. TraceVLA adds visual trajectory prompts to increase

spatial-temporal awareness [26]; SpatialVLA injects 3D spatial representations [17]; OC-VLA predicts actions in camera space to reduce observation-action mismatch [23]; and VistaBot uses geometry-aware view synthesis for cross-view robustness [8]. These methods show that perception is a central bottleneck, but they depend on explicit prompts, 3D encodings, camera coordinate transforms, or generative view modules. A complementary line questions proprioception itself: State-free Policy removes proprioceptive inputs to avoid trajectory overfitting [24], while GAP shows that proprioceptive gradients can suppress visual learning during motion transitions [15]. PROPRIOVLA takes a different stance: it does not discard proprioception or treat it as a passive token. Instead, it uses proprioceptive change to mine token-level self-grounding targets, turning the robot's own motion into self-supervision for learned visual attention.

3 Does a VLA Know Where It Is?

A VLA can recognize the task object while still failing to locate the body that will act on it. We diagnose this missing-self failure with a two-stage probe. First, we expose a behavioral symptom: when the physical manipulation state is unchanged but the wrist-camera view is shifted, the predicted action can shift with the image. Second, we inspect the model's visual reasoning and find that robot-centric queries do not reliably localize the robot body itself. Together, these probes suggest that image-space shortcuts arise because current VLAs can lack visual self-grounding.

3.1 VLAs Can Rely on Image-Space Shortcuts

We first isolate whether the predicted action is anchored to the physical robot-scene state or to the image frame. The robot proprioception, object configuration, language instruction, and ground-truth action are held fixed, while the wrist-camera view is laterally shifted to the left or to the right. This perturbation changes where the scene appears in the image, but it does not change the correct robot-centric action.

A spatially grounded policy should remain stable under this view-only perturbation. Instead, we observe co-directional action drift: left-shifted views induce left-biased trajectories, and right-shifted views induce right-biased trajectories relative to the ground truth. This drift substantially reduces task success, indicating that the policy can complete training-view tasks without fully anchoring action prediction to robot-scene geometry.

This behavior reveals an image-space shortcut. The policy can partially tie control to pixel coordinates, camera-specific correlations, or object-location priors learned from the training view. Under the original camera, these cues can be predictive enough to support imitation. Under a lateral camera shift, however, the same cues move in image space while the correct action remains fixed, causing the action prediction to move with the view. We quantify this effect with Trajectory Shift Ratio (TSR), which measures the magnitude of prediction drift, and Shift Direction Agreement (SDA), which measures whether the drift follows the image shift.

3.2 The Missing Self-Grounding Capability in VLAs

We next ask why image-space shortcuts remain available by probing whether the model visually grounds the robot itself. In both simulation and real-robot scenes, we query the VLM or VLA backbone with robot-centric prompts such as "Where is the robot itself?" and visualize the resulting attention or token relevance maps. The prompt is only a diagnostic tool: because it explicitly names the robot, a visually self-grounded model should assign high relevance to the visible arm, gripper, or end-effector.

The attention maps often fail to concentrate on the robot body. Instead, they assign relevance to task objects, tabletop regions, or other visually salient context, even when the robot is clearly visible. This indicates that task grounding and self-grounding are not the same capability. A VLA may identify the instructed object and still fail to represent where the acting body appears in the image.

This model-level probe supports our attribution of the behavioral failure. Attention maps do not fully explain the policy's internal computation, but they provide consistent evidence that current VLAs can lack visual self-grounding. When the visual pathway does not encode where the robot body appears, the policy can satisfy action supervision through image-space correlations under the training camera.

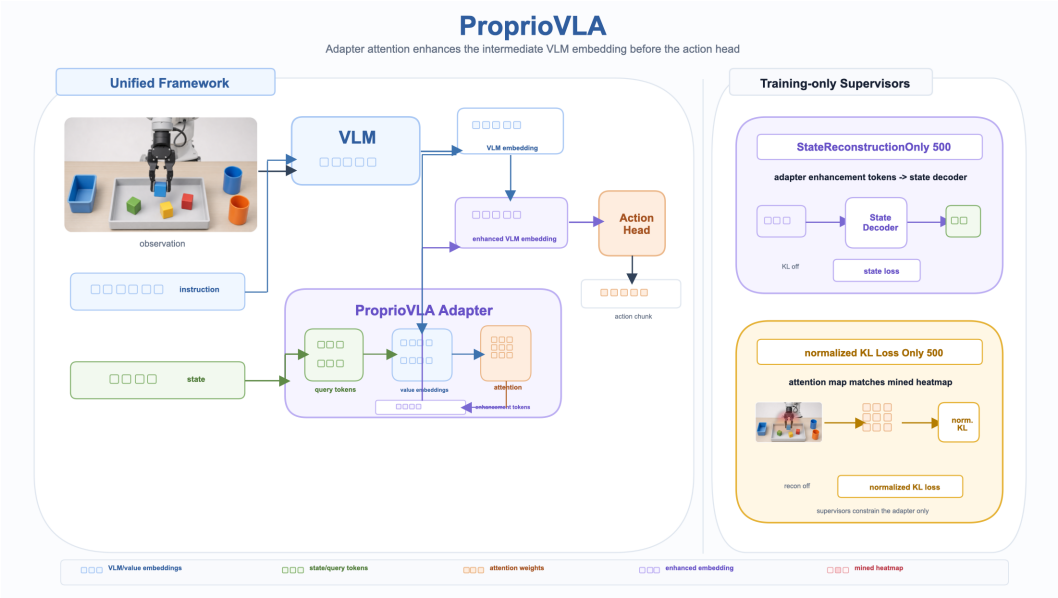

Figure 1: Overview of PROPRIOVLA. Given an observation, instruction, and proprioceptive state, the base VLA produces a standard multimodal representation. State-Aligned Residual Flow (SARF) mines token-level self-grounding targets from proprioceptively aligned visual change, while the Proprioceptive Alignment Module (PAM) trains learned grounding queries over VLA visual tokens to recover these targets. The mined target distribution is used only during training.

Those correlations then become brittle shortcuts under lateral viewpoint shifts. Robust VLA control therefore needs a training signal that binds proprioceptive state to visual evidence of the robot body, rather than assuming that self-grounding emerges from language and action supervision alone.

4 PROPRIOVLA

PROPRIOVLA adds a self-supervised proprioceptive grounding objective to a VLA policy without changing the policy's action interface. Given an observation o_t, language instruction x, proprioceptive state s_t, and action label a_t, the base VLA produces multimodal tokens and is trained with its standard imitation or diffusion action loss. PROPRIOVLA adds two training-time components: State-Aligned Residual Flow (SARF), which mines token-level self-grounding targets from robot interaction, and a Proprioceptive Alignment Module (PAM), which predicts a self-grounding distribution over VLA visual tokens. The pseudo-targets and grounding loss are used only during training; at inference, the policy consumes the same inputs as the base VLA.

Problem Setup and Visual Tokens. We formulate proprioceptive grounding as an auxiliary distribution-matching problem over the VLA visual-token grid. Let the VLA backbone return a sequence of hidden states $Z_t = \{z_{t,i}\}_{i=1}^{L}$. An image-token mask selects the subset of tokens that originate from visual inputs, and the image-token layout records the spatial grid shape for each view. For the target camera view, we collect the valid visual tokens

$$V_t = \{v_t(p) \in \mathbb{R}^d\}_{p \in \mathcal{P}_t},$$

where $\mathcal{P}_t$ indexes valid spatial token locations. Each token has a normalized center coordinate $u_p = (x_p, y_p) \in [0,1]^2$, computed from the corresponding image grid. If multiple image units are present, we group units by view index and apply the grounding objective to the selected view; invalid or missing views are ignored by the token mask.

The objective is to make the VLA representation recover a distribution over where the acting body appears. During training, SARF provides a target distribution H_t over $\mathcal{P}_t$. PAM predicts a distribution A_t from the VLA visual tokens. The grounding loss supervises A_t to match H_t, while gradients flow through the same visual representation used by action prediction. This design separates grounding

supervision from action execution: grounding is learned as a side objective, not as an additional runtime input.

4.2 State-Aligned Residual Flow

SARF mines self-grounding targets from motion that is visually salient and proprioceptively supported. For an episode, we first compute a scalar proprioceptive activity signal from end-effector translation and gripper change,

$$\gamma_t = \mathrm{norm}\big(\|\Delta r_t\|_2 + \beta|\Delta g_t|\big),$$

where r_t is the end-effector position, g_t is the gripper state, β balances gripper activity, and $\mathrm{norm}(\cdot)$ denotes robust percentile normalization. High γ_t indicates frames where the robot state changes enough to make visual self-motion informative.

SARF next estimates residual visual change from consecutive frames. Let I_t be the grayscale image at time t, and let $F_{t\to t+1}$ be dense optical flow between adjacent frames. We combine temporal image difference and optical-flow magnitude into a visual-change score

$$M_t = \mathcal{N}\left(\frac{1}{2}\left(|I_t - I_{t-1}| + |I_{t+1} - I_t|\right) + \frac{1}{4}\left(\|F_{t-1\to t}\|_2 + \|F_{t\to t+1}\|_2\right)\right),$$

where $\mathcal{N}$ denotes map normalization followed by mild Gaussian smoothing. Image difference captures appearance changes, while optical flow captures coherent displacement. Their combination suppresses static background tokens and highlights image regions whose change can plausibly correspond to the robot or objects directly coupled to it.

To reduce spurious motion from distractors, SARF fits a state-guided spatial prior from high-confidence motion peaks within each episode. A lightweight weighted regression maps proprioceptive state features to image points, using motion-derived confidence as weights and robust inlier support to reject outliers. When the fit is reliable, it predicts a prior point μ_t and support score q_t for each frame. SARF combines this prior with visual change as

$$S_t(p) = \mathcal{N}\left(M_t(p)^{0.85}\left[(1-\rho_t) + \rho_t G(p;\mu_t,\sigma_{\mathrm{prior}})\right]\right),$$

where G is a Gaussian point prior and ρ_t is capped by the fitted model quality. If the state-guided fit is unreliable, SARF falls back to the pure motion score M_t. This fallback keeps target mining self-supervised and avoids forcing a poor state-to-image fit onto ambiguous episodes.

SARF converts the score map into a token-level target. We blur S_t, extract a peak region from the high-score set, and obtain a normalized pseudo point y_t, anisotropic scale $\sigma_t = (\sigma_t^x, \sigma_t^y)$, bounding box B_t, and confidence c_t. The confidence is a geometric combination of peak sharpness, visual-change energy, proprioceptive activity, and, when available, state-prior support. Low-confidence frames are abstained from by setting $c_t = 0$ below a minimum threshold.

Given token centers $\{u_p\}_{p\in\mathcal{P}_t}$, we rasterize the mined pseudo point into a normalized target distribution:

$$\tilde{H}_t(p) = \exp\left(-\frac{1}{2}\left[\left(\frac{u_p^x - y_t^x}{\sigma_t^x}\right)^2 + \left(\frac{u_p^y - y_t^y}{\sigma_t^y}\right)^2\right]\right)\mathbb{1}[u_p \in B_t],$$

$$H_t(p) = \frac{\tilde{H}_t(p)}{\sum_{q\in\mathcal{P}_t}\tilde{H}_t(q) + \epsilon}.$$

If the clipped box contains no valid token, we fall back to the unclipped Gaussian target, and if that also has no mass, to the uniform distribution over valid tokens. Thus SARF produces a stable token-level self-grounding target from interaction alone, without robot-part masks or external grounding labels.

4.3 Proprioceptive Alignment Module

PAM predicts the self-grounding distribution with a lightweight learned-query head over VLA visual tokens. The module maintains M trainable grounding latents $Q = \{q_m\}_{m=1}^{M}$. In the main learned-query setting, these latents are independent parameters; a state-conditioned query variant, used for ablation, adds a projection of the grounding state to the same latents. Each visual token is augmented with a learned positional embedding of its normalized center coordinate u_p.

PAM applies a small stack of cross-attention blocks from grounding latents to visual tokens. For one block, with token inputs $\hat{v}_t(p)$, the attention logits are

$$\ell_t(m,p) = \bar{q}_m^\top \bar{k}_t(p), \qquad \bar{q}_m = \mathrm{norm}(W_q\,\mathrm{LN}(q_m)), \quad \bar{k}_t(p) = \mathrm{norm}(W_k\,\mathrm{LN}(\hat{v}_t(p))).$$

The block uses the corresponding softmax attention to update the grounding latents through value projection and a feed-forward layer. The final-layer attention logits define the predicted self-grounding distribution:

$$A_t(p) = \frac{1}{M}\sum_{m=1}^{M} \mathrm{softmax}_{q\in\mathcal{P}_t}\big(\ell_t(m,q)\big)_p.$$

Averaging across latents lets several queries cover the same visible embodiment while still producing one distribution for supervision.

PAM is intentionally a side head rather than a token-writeback module. The grounding branch reads VLA visual tokens and returns attention logits for training, but the original VLA token sequence and attention mask are passed to the action decoder unchanged. In the normalized-KL-only setting, state reconstruction and feature reconstruction auxiliaries are disabled, and the grounding loss acts through the attention-producing query/key pathway. This keeps the method faithful to the central goal: shaping shared visual tokens to encode visible embodiment, without adding pseudo-label-dependent inputs to the policy at test time.

Normalized KL Grounding Objective. The grounding objective compares the predicted distribution A_t to the SARF target H_t with a scale-normalized KL divergence. Let

$$\mathrm{CE}_t = -\sum_{p\in\mathcal{P}_t} H_t(p)\log A_t(p), \qquad \mathcal{H}(H_t) = -\sum_{p\in\mathcal{P}_t} H_t(p)\log H_t(p).$$

The KL from target to model is $\mathrm{CE}_t - \mathcal{H}(H_t)$. Because its magnitude depends on token count and target sharpness, we normalize it by the KL from the target to uniform attention:

$$\kappa_t = \frac{\mathrm{CE}_t - \mathcal{H}(H_t)}{\log|\mathcal{P}_t| - \mathcal{H}(H_t) + \epsilon}.$$

With this normalization, a target-matching prediction approaches zero, while uniform attention is approximately one. This makes the auxiliary loss stable across different image-token counts, pseudo-box sizes, and target sharpness.

The complete normalized-KL-only training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{act}} + \lambda_g \frac{\sum_t \mathbb{1}[c_t \geq c_{\min}] c_t \kappa_t}{\sum_t \mathbb{1}[c_t \geq c_{\min}] c_t + \epsilon}.$$

$\mathcal{L}_{\mathrm{act}}$ is the original VLA action loss. The implementation keeps $\mathcal{L}_{\mathrm{act}}$ active in all grounding modes; in normalized-KL-only mode, it forces the localization auxiliary to use normalized KL and sets the state-reconstruction and feature-reconstruction weights to zero. Since normalized KL is already calibrated against the uniform baseline, it is not further rescaled by action-loss balancing. The result is a training signal that teaches visual self-grounding while preserving the standard VLA inference path.

5 Experiments

The experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. The evaluation links three evidence types: task performance, action stability under camera perturbations, and visual grounding of the acting robot.

5.1 Experimental Setup

We evaluate a VLA backbone fine-tuned on language-conditioned manipulation demonstrations in simulation and on a real robot. Each episode contains visual observations o_t, proprioceptive state s_t, language instruction x, and action supervision a_t. For robustness evaluation, we keep the physical robot-scene state, instruction, and proprioceptive state fixed, then apply lateral view shifts

of controlled magnitude. This isolates camera-induced changes from changes in the underlying manipulation state.

We compare four policies under the same demonstrations, action labels, and view-shift protocol: Base VLA, action-supervised fine-tuning, VLA with proprioception concatenation, and PROPRIOVLA. We report task success rate, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and a grounding score. The grounding score is computed as either attention-target alignment with H_t or attention mass on robot regions. Lower trajectory shift and direction agreement indicate a more stable action frame; higher grounding score indicates stronger localization of the acting body. Each setup reports the task suite, demonstration count, train/test split, camera shifts, action parameterization, proprioceptive normalization, optimizer, and the grounding loss weight λ_g.

5.2 Simulation Experiments

Simulation provides controlled access to camera shifts, object layouts, and ground-truth robot geometry. We evaluate normal, mild-shift, medium-shift, and large-shift settings. For each setting, we report task success, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and grounding score. Robustness curves plot these metrics against shift magnitude, making the co-movement between predicted actions and image shifts directly visible.

We also evaluate held-out object layouts and camera positions. These splits separate memorization of a fixed training camera from grounding that transfers across views. Qualitative trajectory overlays compare the original prediction with shifted-view predictions after mapping them to the same robot-centric coordinate frame.

5.3 Real-World Experiments

The real-robot evaluation tests whether the same failure mode appears under sensor noise, calibration error, and real scene variation. We evaluate several manipulation tasks under the original view and shifted views, reporting task success, success drop under perturbation, action prediction error on logged demonstrations, and grounding score.

For qualitative analysis, we visualize baseline attention, the mined state-aligned target distribution, PROPRIOVLA attention, and trajectory overlays for the same scene. The comparison checks whether attention remains dominated by task objects and background context or shifts toward the arm, gripper, and end-effector while the trajectory remains stable.

5.4 Ablations

The ablation study isolates the mechanisms that distinguish PROPRIOVLA from simple state conditioning. We evaluate four variants: (1) removing the visual self-grounding loss $\mathcal{L}_{\text{ground}}$; (2) replacing normalized KL with an unnormalized localization loss; (3) using raw optical flow instead of state-aligned residual motion; and (4) replacing the learned grounding queries with state-derived queries. Each ablation is evaluated with the same success, action, trajectory-shift, and grounding metrics.

These ablations test the central mechanism: the policy must use proprioception to mine visual self-grounding targets, rather than merely receive proprioceptive values as extra inputs. The comparison separates three alternatives: gains from additional state, gains from motion saliency alone, and gains from learned-query self-grounding.

6 Limitations

PROPRIOVLA assumes that the robot is at least partially visible during training. If the embodiment is consistently outside the camera view, the state-aligned motion signal cannot provide useful self-grounding supervision, and the visual query has no direct evidence from which to recover the robot's own state.

State-aligned residual motion can be ambiguous when non-robot motion is coupled to the robot. Grasped objects, deformable materials, moving backgrounds, or other agents may move synchronously with proprioceptive change and receive high alignment scores. Confidence weighting

reduces weak supervision, but it cannot fully distinguish the robot body from every object moved by that body.

Finally, the evaluation covers a finite set of embodiments, camera perturbations, and manipulation tasks. Broader multi-camera setups, mobile manipulation, multi-robot scenes, long-horizon tasks, and highly dynamic environments remain important tests for whether proprioceptive self-grounding scales beyond controlled perturbation regimes.

7 Conclusion

This paper studies a missing-self failure mode in VLA policies: under camera-only viewpoint shifts, predicted robot trajectories can move with the image frame even when the physical robot-scene state is unchanged. The diagnostic shows that task grounding and self-grounding are distinct requirements for spatially reliable manipulation.

PROPRIOVLA addresses this issue through two connected components: State-Aligned Residual Flow (SARF) and a Proprioceptive Alignment Module (PAM). SARF mines token-level self-grounding targets from robot interaction without manual masks or external grounding datasets, and PAM trains learned grounding queries to recover these targets with normalized KL supervision. Learning to see the self is a practical step toward VLA policies whose actions are grounded in embodiment rather than in the image frame alone.

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
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